(Please write your Enrollment Number) Enrollment No. 0160117202

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(Offline)
Course Name: B. Tech CSE, CSE-AI, IT>, Semester:4th
(April -May, 2023)

Subject Code: BIT-202	Subject: Operating Systems
Time :3 Hours	Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

(4*5=20)

Q1

(a) How does RAID structure guarantee improvement of reliability and performance?

(b) When does context switch occur? Describe the actions taken by the OS when a context switch occurs.

(c) When does race condition arise? Can we guard against this condition? How?

(d) How can paging and segmentation be combined together?

UNIT-I

Q2 Consider the following set of processes with the CPU burst & arrival time given in milliseconds: (10)

Process	Arrival Time	Burst Time	Priority
P1	0	9	3
P2	1	1	1
P3	2	2	3
P4	3	4	4
P5	4	5	2

Draw the Gantt chart & calculate average waiting time & average turnaround time for each of the following scheduling algorithms:

i) Shortest Job First
ii) Shortest Remaining Time First
iii) Round Robin (Time Quantum=3ms) scheduling algorithm.
iv) Non Preemptive priority scheduling

Q3 Consider the following set of processes with the length of CPU burst time given in milliseconds: (10)

Process	Arrival Time	Burst Time	Priority
P1	4	10	4
P2	3	1	1
P3	2	2	4
P4	1	3	5
P5	0	5	3

PTO

Draw the Gantt chart & calculate the average waiting time & average turnaround time for the following scheduling algorithms:

i) Shortest Job First, ii) Shortest Remaining Time First, iii) Round Robin (Time Quantum=3ms), iv) Non Preemptive Priority scheduling.

UNIT-II

Q4 Consider there are 3 frames allocated to a process and the reference string is: 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3. Calculate the number of page faults for the following page replacement algorithms: (10)

(i) FIFO (ii) LRU (iii) Optimal (iv) LFU (v) MFU

Q5 Consider there are 4 frames allocated to a process and the reference string is: 4, 3, 2, 1, 4, 3, 5, 4, 3, 2, 1, 5. Calculate the number of page faults for the following page replacement algorithms: (10)

(i) FIFO (ii) LRU (iii) Optimal (iv) LFU (v) MFU

UNIT-III

Q6 Consider the following snapshot of the system: (10)

Process	Allocation A B C D	Max A B C D	Available A B C D
P0	0 0 1 2	0 0 1 2	1 5 2 0
P1	1 0 0 0	1 7 5 0	
P2	1 3 5 4	2 3 5 6	
P3	0 6 3 2	0 6 5 2	
P4	0 0 1 4	0 6 5 6	

(i) What is the content of the NEED matrix?

(ii) Is the system in a safe state? Explain.

Given the current snapshot of the system, if a request from P1 arrives for (0, 4, 2, 0), can the request be granted immediately?

Q7 Suppose a disk drive has 3000 cylinders (0 to 2999). The drive is currently serving a request at cylinder 125 and the previous request was at cylinder 145. The queue of pending request is 233, 456, 125, 91, 789, 1600, 2345, 13. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the requests? (10)

1. Elevator 2. LOOK 3. C-SCAN 4. C-LOOK

UNIT-IV

Q8 What is a directory? What are the different ways of implementing a directory? Explain the allocation schemes that exist for allocating secondary storage to files. (10)

Q9 Write short notes on the following: (10)

(i) Disk management
(ii) Free space management
(iii) Swap space management
(iv) Disk Structure